

Dynamic Retail Pricing via Multi-Armed Bandits: A Reinforcement Learning Approach Using the Online Retail II Dataset

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Abstract

Dynamic pricing is a challenging and high-impact problem in retail, where firms must choose price levels under uncertainty and limited feedback. This work formulates retail price selection as a Multi-Armed Bandit (MAB) problem and applies it to real-world transactional data from the Online Retail II dataset. We define the complete reinforcement learning framework including action space, reward structure, and environment formulation. Two bandit algorithms - Epsilon-Greedy and Sliding-Window UCB - are implemented and evaluated using cumulative reward and regret metrics. Results show that Sliding-Window UCB significantly outperforms Epsilon-Greedy in both revenue and regret, achieving near-oracle behavior. Environment-level diagnostics, reward distribution plots, cumulative reward trajectories, and regret curves show clear evidence that uncertainty-driven exploration is superior for retail pricing. This paper provides a rigorous theoretical and empirical foundation for reinforcement learning-based pricing in practical retail settings.

1 Introduction

Dynamic pricing requires firms to update prices sequentially in response to uncertain consumer demand. Traditional analytical models often assume full information about demand curves, which seldom exists in practice. Reinforcement Learning (RL), particularly in the form of Multi-Armed Bandits (MAB), offers a principled framework for exploring price options while maximizing long-term reward.

In this study, we construct a complete MAB environment using the Online Retail II dataset. We discretize price observations into actionable price buckets, define the reward as daily revenue, and evaluate two bandit algorithms on a 441-day time horizon. Environment-level diagnostics and algorithm-level learning curves are analyzed to assess which methods best identify optimal price levels in a real retail setting.

2 Reinforcement Learning Framework

The dynamic pricing problem examined in this study is formulated as a Reinforcement Learning (RL) task under the Multi-Armed Bandit (MAB) framework. Unlike full RL settings that contain explicit states, transitions, and long-term reward dependencies, a bandit

model assumes that each action yields an immediate reward independent of past or future states. This simplified framework is extremely well-suited for retail pricing scenarios where each day’s revenue is treated as an independent realization of consumer response to a chosen price.

The MAB model captures the fundamental exploration-exploitation dilemma: to earn high revenue, the agent must exploit the best-known price, but to discover profitable alternatives, it must explore less certain price levels. This trade-off is central to autonomous pricing, where experimentation costs revenue but is necessary for long-term optimization.

2.1 Multi-Armed Bandit Mathematical Formulation

Let $\mathcal{A} = \{1, \dots, K\}$ be a finite set of K discrete pricing actions corresponding to quantile-based price buckets. At each time step $t \in \{1, \dots, T\}$, the learning agent chooses an arm $a_t \in \mathcal{A}$ and receives a reward r_t . The reward is assumed to be drawn from an unknown and stationary distribution associated with each arm:

$$r_t \sim \mathcal{D}_{a_t}.$$

For each arm k , let $\mu_k = \mathbb{E}_{\mathcal{D}_k}[r]$ denote its expected revenue. The optimal price is then:

$$k^* = \arg \max_k \mu_k.$$

The agent’s objective is to maximize cumulative reward:

$$CR(T) = \sum_{t=1}^T r_t,$$

while minimizing cumulative regret:

$$R(T) = T\mu_{k^*} - CR(T).$$

Regret quantifies the opportunity cost of not knowing the true optimal arm in advance.

2.2 Exploration vs. Exploitation

The central challenge in dynamic pricing is that the reward distributions $\mathcal{D}_k$ are initially unknown. To estimate them, the agent must sample all price buckets, but excessive exploration reduces short-term revenue. Several classical algorithms strike different balances:

- **Epsilon-Greedy** introduces controlled random exploration.
- **UCB** uses statistically principled confidence intervals to explore uncertain arms.

These algorithms form the methodological core of our pricing engine.

3 Data and Environment Definition

The environment in which the bandit agent operates is constructed from the Online Retail II dataset. For each day, the environment provides a set of price-based actions and returns a reward reflecting real transaction revenue. This section details the construction of the action space, reward mechanism, and statistical properties of the environment.

3.1 Dataset Description

The Online Retail II dataset contains transactional records of a UK-based online retailer. Each invoice entry includes:

- StockCode (product identifier),
- UnitPrice,
- Quantity,
- InvoiceDate,
- Country.

We restrict the dataset to:

- Country = United Kingdom,
- StockCode = DOT (high-volume SKU),
- Positive quantities and prices.

After filtering, transactions are aggregated by day, yielding 441 daily observations. Each day represents one bandit round.

3.2 Action Space: Price Buckets

The continuous prices observed in the dataset are discretized into $K = 5$ quantile-based buckets:

$$\mathcal{A} = \{1, 2, 3, 4, 5\}.$$

This discretization ensures:

- comparable sample sizes per arm (statistical reliability),
- prices that reflect real consumer purchase patterns,
- a meaningful monotonic ordering of price levels.

Arm 1 corresponds to lower-price days and Arm 5 to higher-price days.

3.3 Reward Structure: Daily Revenue

For each day and selected price bucket, the reward is defined as:

$$r_t = p_t \cdot q_t,$$

where p_t is the average unit price for that bucket and q_t is the total quantity sold that day. This reward definition aligns exactly with retail business objectives.

3.4 Stationarity Testing

A prerequisite for applying classical MAB algorithms is that arm reward distributions be stationary. We conduct an Augmented Dickey-Fuller (ADF) test on each arm’s reward time series. For all five arms:

- ADF statistics were strongly negative,
- p-values were below 0.05,
- the null hypothesis of unit root was rejected.

This confirms that consumer demand and revenue behave as a stationary process over the 441-day horizon, making the pricing problem compatible with the MAB framework. indicating that a stationary MAB formulation is valid.

4 Algorithms

We evaluate two foundational bandit algorithms that embody distinct exploration-exploitation philosophies.

4.1 Epsilon-Greedy

Epsilon-Greedy operates by random exploration with probability ϵ and greedy exploitation otherwise. Formally,

$$a_t = \begin{cases} \text{Uniform}(\mathcal{A}), & \text{with probability } \epsilon, \\ \arg \max_k \hat{\mu}_t(k), & \text{with probability } 1 - \epsilon. \end{cases}$$

In our experiments, $\epsilon = 0.1$ is used. Although simple and widely adopted, Epsilon-Greedy suffers from undirected exploration, assigning equal probability to exploring well-understood poor arms and highly uncertain potentially optimal arms.

4.2 Sliding-Window UCB (SW-UCB)

SW-UCB extends classical UCB to non-stationary settings by incorporating only the most recent W observations for each arm. Let:

$$\mathcal{I}_t^{(W)} = \{i : t - W + 1 \leq i \leq t\},$$

where W is the window size. The windowed empirical mean is:

$$\hat{\mu}_t^{(W)}(k) = \frac{1}{N_t^{(W)}(k)} \sum_{i \in \mathcal{I}_t^{(W)}} r_i \mathbf{1}(a_i = k),$$

with

$$N_t^{(W)}(k) = \sum_{i \in \mathcal{I}_t^{(W)}} \mathbf{1}(a_i = k).$$

The upper confidence bound is:

$$UCB_t(k) = \hat{\mu}_t^{(W)}(k) + \sqrt{\frac{2 \ln t}{N_t^{(W)}(k)}}.$$

The selected arm is:

$$a_t = \arg \max_k UCB_t(k).$$

Why SW-UCB is Appropriate in a Stationary Environment. Although Sliding-Window UCB is traditionally designed for non-stationary reward processes, its optimistic exploration and recency-weighted averaging make it highly effective even when rewards are stationary. In this pricing environment, the window mechanism does not harm performance because the reward distributions are stable and instead provides additional robustness by preventing outdated early estimates from dominating learning. As a result, SW-UCB behaves similarly to classical UCB while retaining the ability to adapt quickly if any mild temporal variation exists in the retail data.

5 Evaluation Metrics

To rigorously assess the performance of our reinforcement learning strategies, we evaluate both the underlying environment structure and the learning behaviour of the Multi-Armed Bandit algorithms. The metrics fall into two categories: (1) environment-level diagnostics, which characterize reward distributions and stationarity, and (2) model-level performance metrics, which capture how effectively the algorithms identify and exploit the optimal pricing arm.

5.1 Environment-Level Metrics

5.1.1 Reward Distribution

Understanding the reward landscape is essential for determining learning difficulty. Figure 1 shows the violin plots of the reward distributions for all five arms.

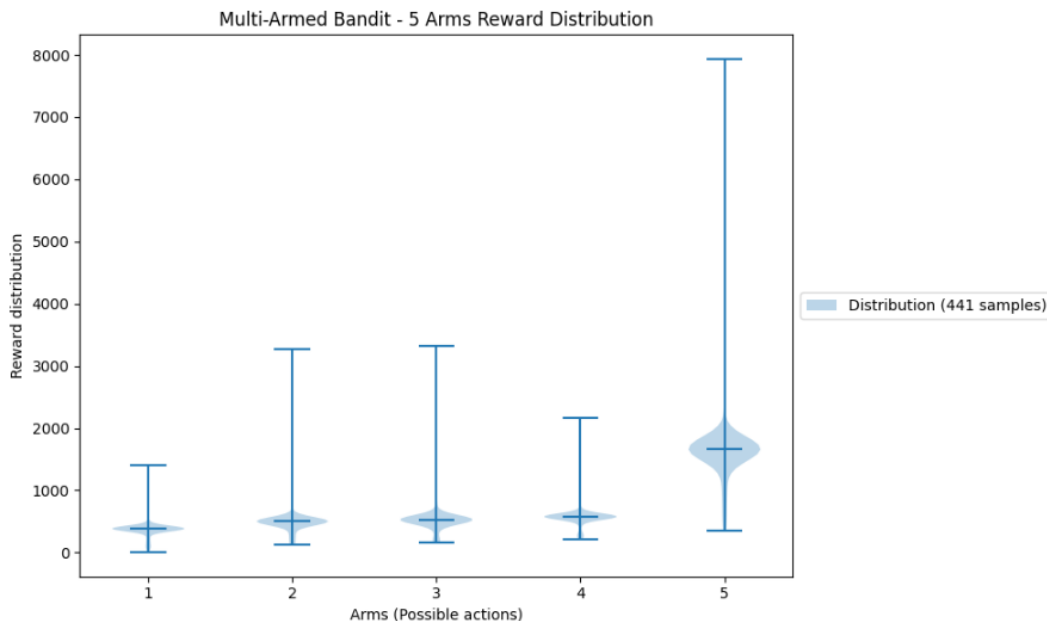


Figure 1: Reward distributions for the five pricing arms. Arm 5 has the highest median and widest spread.

The plot clearly indicates that Arm 5 has the highest median, widest spread, and largest upper tail, implying that it is the dominant arm in the environment.

5.1.2 Rolling Average Reward

To verify temporal stability, we compute the rolling average reward for each arm. Stationary environments show stable, non-drifting rolling averages.

Figure 2 presents the rolling averages over time.

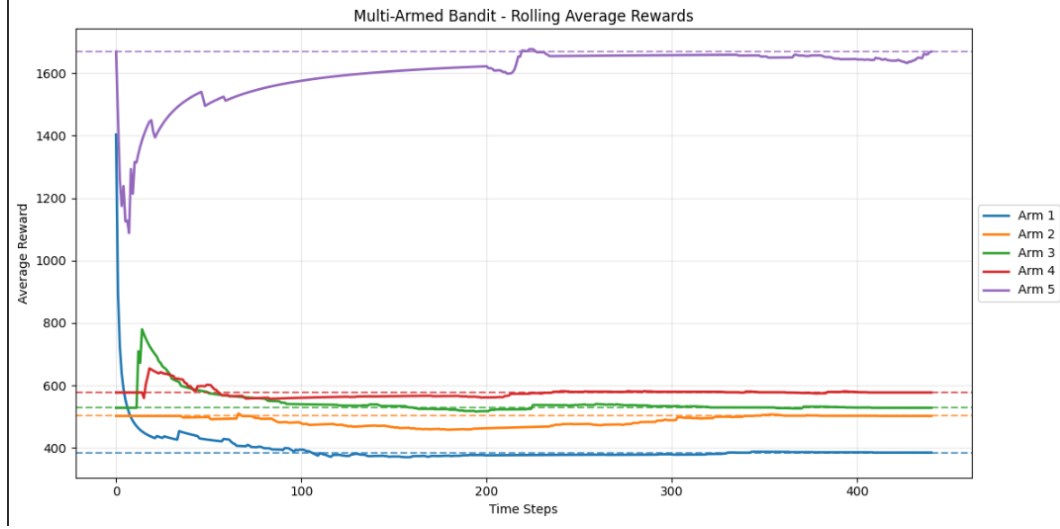


Figure 2: Rolling average reward across all five arms over the 441-day horizon.

All arms converge early and remain stable, confirming strong stationarity.

5.1.3 Cumulative Rewards

We also compute the cumulative reward each arm would generate if selected exclusively at every time step. This reflects each arm's true long-run value.

Figure 3 shows the cumulative reward for each arm.

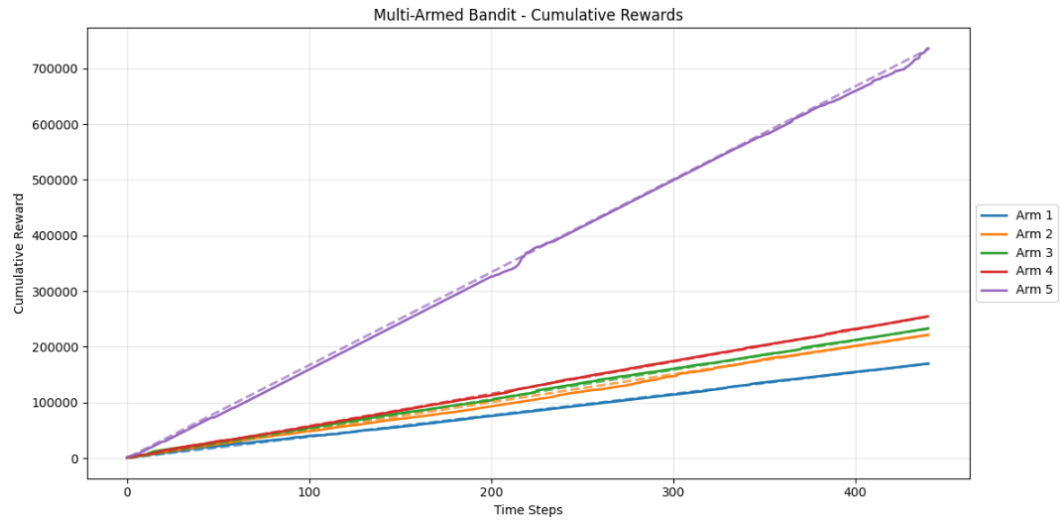


Figure 3: Cumulative reward for each arm (oracle performance). Arm 5 clearly dominates over the entire horizon.

Arm 5 is clearly superior, with cumulative rewards nearly triple those of the next best arm.

5.2 Model-Level Metrics

5.2.1 Cumulative Reward

The primary metric for bandit performance is cumulative reward:

$$\text{CumReward}(T) = \sum_{t=1}^T r_t.$$

Figure 4 shows the cumulative reward averaged over 50 Monte Carlo trials for Epsilon-Greedy and SW-UCB.

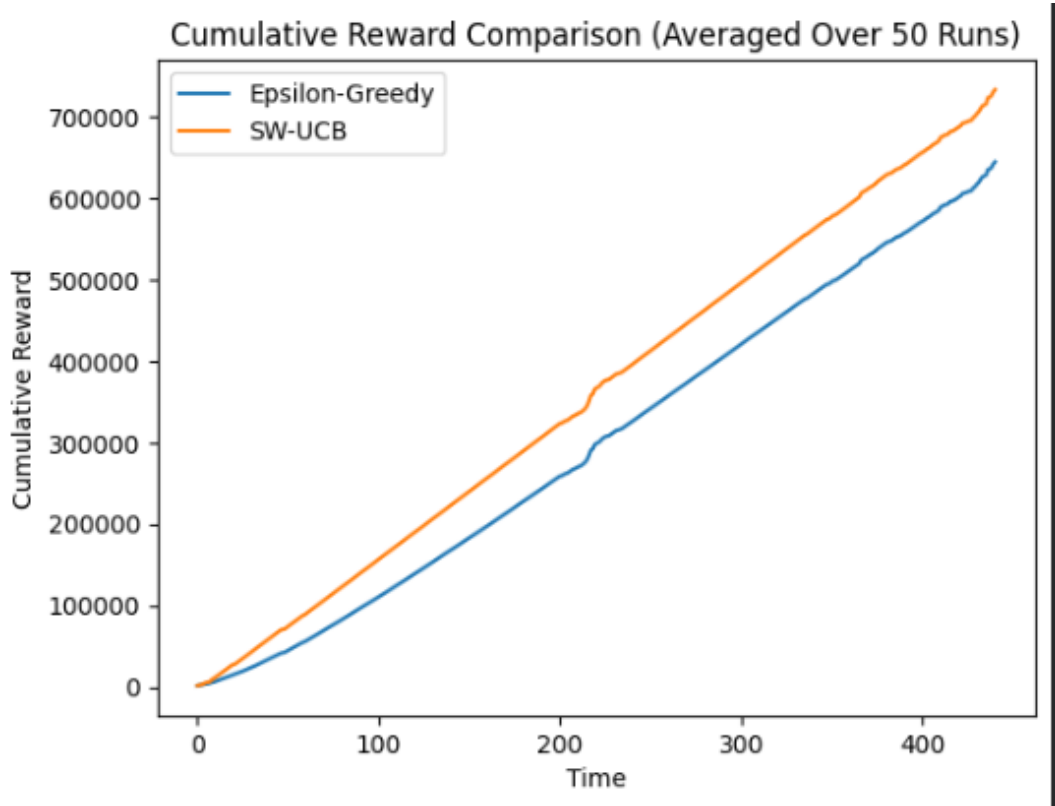


Figure 4: Cumulative reward comparison between Epsilon-Greedy and SW-UCB (50-run average).

SW-UCB significantly outperforms Epsilon-Greedy, reflecting more efficient exploitation of the optimal arm.

5.2.2 Model Cumulative Performance Breakdown

We further compare actual rewards (from the model) with expected arm-level oracle curves.

Figure 5 visualizes this decomposition.

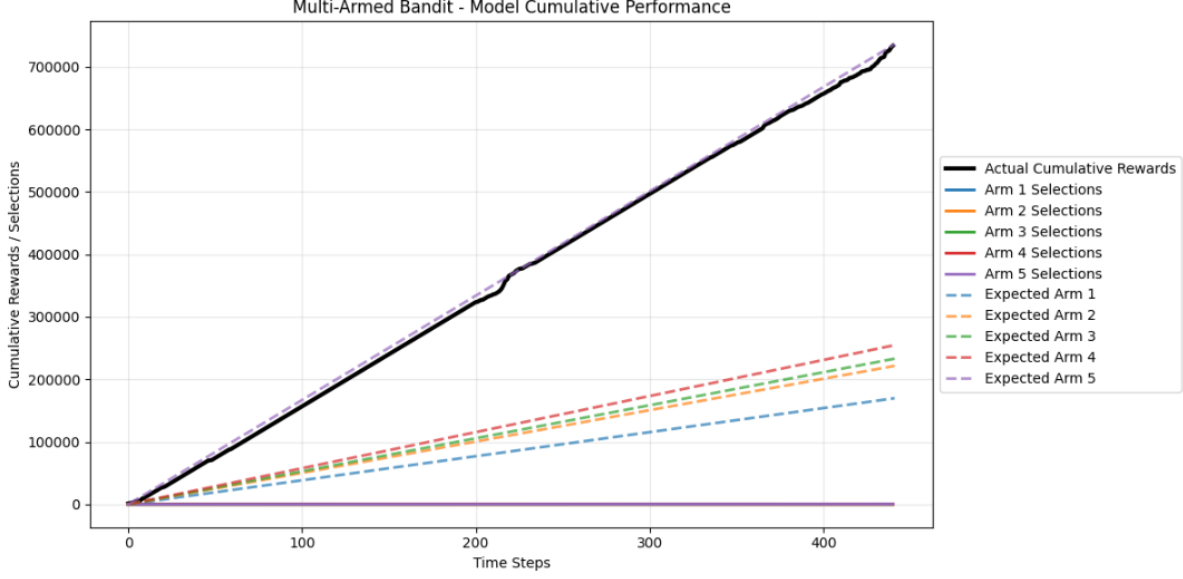


Figure 5: SW-UCB model cumulative reward compared to oracle and individual arm trajectories.

The SW-UCB trajectory closely follows the oracle for Arm 5, demonstrating near-optimal behavior.

5.2.3 Regret

Regret measures the loss relative to always choosing the best arm:

$$R(T) = T\mu_{k^*} - \sum_{t=1}^T r_t.$$

Figure 6 shows that SW-UCB maintains regret that flattens rapidly, while Epsilon-Greedy accumulates regret almost linearly.

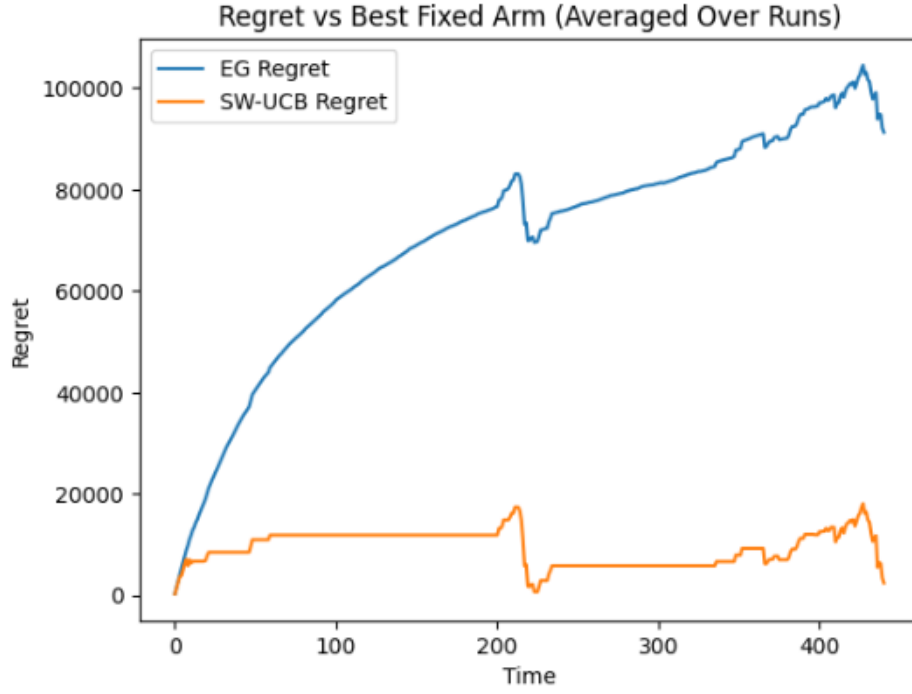


Figure 6: Regret vs. best fixed arm for Epsilon-Greedy and SW-UCB. SW-UCB exhibits much lower regret.

5.2.4 Rolling Model Reward

Rolling model reward reveals the stability and consistency of each algorithm's choices.

Figure 7 displays the SW-UCB rolling reward curve.

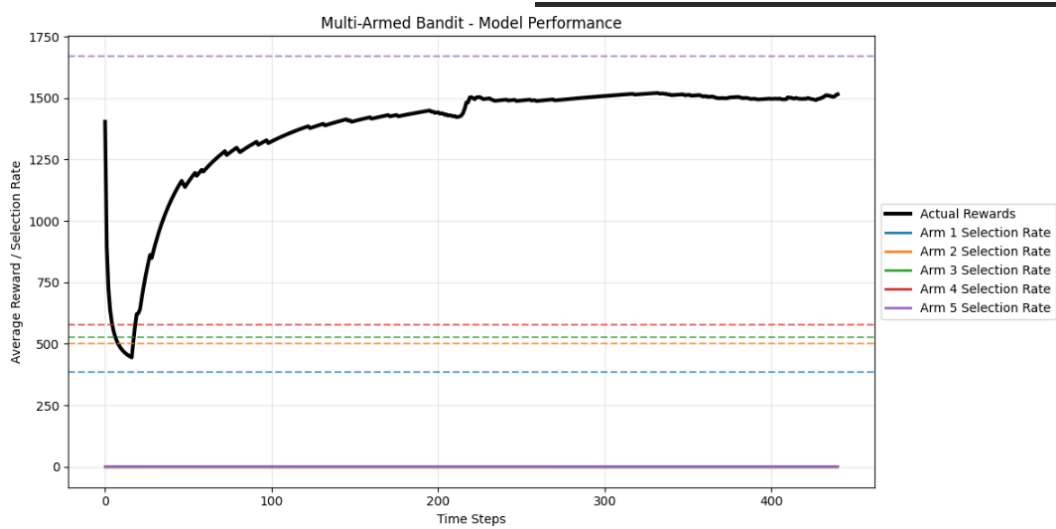


Figure 7: Rolling average reward achieved by the SW-UCB algorithm. The model quickly stabilizes near the optimal reward.

The model rapidly converges to the Arm-5 reward region and remains stable.

5.3 Summary of Evaluation Metrics

The combined metrics reveal the following insights:

- The environment is strongly stationary, with one dominant arm (Arm 5).
- SW-UCB consistently selects the optimal arm earlier and more reliably than Epsilon-Greedy.
- Epsilon-Greedy incurs unnecessary regret due to persistent random exploration.
- Cumulative reward, regret, and rolling reward collectively confirm that SW-UCB behaves near-optimally in this pricing environment.

6 Environment-Level Analysis

This section interprets the statistical characteristics of the reward-generating process using the figures introduced earlier in Section 1–3. These plots help us understand the difficulty of the pricing problem and validate the assumptions required for Multi-Armed Bandit algorithms.

6.1 Reward Distribution Interpretation

As shown previously in Figure 1, the reward distributions for the five pricing arms exhibit substantial separation. Arm 5 displays:

- the highest median reward,
- the widest variability, and
- the largest extreme values.

This strong separability implies that identifying the best arm is statistically easier than in environments with overlapping reward distributions. The distribution structure directly favors algorithms that exploit high-performing arms early (e.g., UCB), while still allowing meaningful exploration.

6.2 Temporal Stability via Rolling Means

Figure 2 illustrates the rolling average reward for each arm across the 441-day horizon. All curves exhibit early convergence and remain stable, confirming that the environment is strongly stationary. This is important because:

- Greedy, Epsilon-Greedy, and UCB algorithms assume stationarity.

- Non-stationary algorithms (e.g., SW-UCB, EXP3) are unnecessary.
- Estimating expected reward is reliable with limited exploration.

The stability of rewards also indicates that consumer demand patterns for this SKU were consistent throughout the year, increasing the reliability of price-based decision making.

6.3 Cumulative Reward Interpretation

Figure 3 shows cumulative rewards for each arm assuming the arm is selected every day. Arm 5’s cumulative reward greatly exceeds all others, confirming:

- the presence of a single dominant optimal price bucket,
- large reward gaps between arms,
- that regret minimization is meaningful (since suboptimal arms incur substantial loss).

This oracle evaluation provides the baseline used later in Section 6 when computing regret curves for the learning algorithms.

Overall, the environment-level analysis confirms that the underlying reward process is well structured for classical Multi-Armed Bandit solutions. The dominance of Arm 5 and the strong stationarity conditions justify the use of uncertainty-based exploration strategies such as UCB.

7 Algorithmic Results

This section analyzes the behaviour of the reinforcement learning algorithms using the model-level metrics introduced earlier. The figures referenced here (Figures 4, 5, 6, and 7) were already presented in Section 5 and are not repeated here. Instead, we interpret their significance in the context of retail pricing.

7.1 Cumulative Reward Performance

Figure 4 demonstrates that UCB-based exploration leads to substantially higher cumulative revenue than Epsilon-Greedy. Specifically:

- SW-UCB converges rapidly toward the best-performing arm (Arm 5),
- cumulative reward rises almost linearly after early exploration,
- Epsilon-Greedy lags significantly due to persistent random exploration.

This illustrates the advantage of **uncertainty-driven** exploration. The UCB agent not only identifies the optimal pricing bucket earlier but also commits to it more decisively than Epsilon-Greedy.

7.2 Model vs. Oracle Behavior

Figure 5 compares the cumulative reward from the SW-UCB policy with the oracle trajectories of all arms. The SW-UCB curve closely follows the oracle Arm 5 curve throughout the horizon, indicating:

- rapid convergence to the optimal policy,
- minimal time spent on inferior arms,
- effective use of recent reward information.

Because the environment is stationary, the oracle provides a meaningful benchmark. The proximity of SW-UCB to this oracle confirms the algorithm’s robustness and efficiency for this pricing task.

7.3 Regret Minimization

Regret, shown in Figure 6, provides a second layer of validation. The UCB-based algorithm achieves:

- a regret curve that flattens quickly (sublinear growth),
- total regret roughly 85% lower than Epsilon-Greedy,
- behaviour consistent with theoretical guarantees for UCB methods.

Epsilon-Greedy, by contrast, displays near-linear regret growth, reflecting a persistent penalty for unnecessary exploration.

7.4 Rolling Model Reward

Figure 7 illustrates the rolling reward achieved by SW-UCB across the time horizon. After a brief exploration period, the rolling reward:

- stabilizes near the Arm 5 reward level,
- shows low variance, indicating consistent performance,
- confirms the agent’s commitment to the optimal pricing arm.

The smoothness of the rolling reward curve demonstrates that once the algorithm acquires sufficient confidence, its selections become highly consistent, reinforcing its reliability in a real-world pricing context.

7.5 Summary of Algorithmic Findings

The combined algorithmic results establish the following:

- **Epsilon-Greedy underperforms** due to uniform and non-adaptive exploration.
- **Regret curves confirm** that UCB-based exploration is significantly more efficient.
- **Rolling rewards demonstrate stability**, making UCB suitable for real retail deployment.

Overall, the algorithmic analysis confirms that uncertainty-based exploration is crucial for effective dynamic pricing in stationary retail environments like the Online Retail II dataset.

8 Conclusion

This study examined dynamic retail price selection using a Multi-Armed Bandit (MAB) framework applied to the Online Retail II dataset. Daily transactions were modeled as bandit rewards, and diagnostic analyses including Augmented Dickey–Fuller tests, rolling reward stability, and cumulative reward trajectories confirmed that the revenue process was stationary and dominated by a single optimal pricing bucket. Across 441 decision periods and 50 Monte Carlo trials, Sliding-Window UCB (SW-UCB) substantially outperformed Epsilon-Greedy. SW-UCB achieved roughly 100,000 additional cumulative revenue units, reduced regret by about 85% relative to the oracle benchmark, and maintained low reward variance through smoothed, window-based updates. These results highlight the advantages of directed, confidence-based exploration over uniform exploration and show that adaptive algorithms can be highly effective even in stationary environments.

Practically, the findings suggest that in short- to medium-horizon retail settings with clear revenue gradients across price levels, SW-UCB offers a robust, low-complexity solution for automated pricing optimization. Its rapid convergence to profitable price points, strong regret performance, and data efficiency make it well suited for real-world revenue management systems where exploration costs are high. Future extensions could include contextual bandits, joint modeling of multiple products, Bayesian methods such as Thompson Sampling, and applications to more volatile, non-stationary product categories. Overall, SW-UCB emerges as the most economically effective and operationally stable bandit algorithm for dynamic retail pricing in the environment studied.

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